

DETERMINANTS

- Any $n \times n = n^2$ numbers arranged in a square form along horizontal rows and vertical columns having n elements in each row and column enclosed between two vertical bars is called a determinant of order n .
- $D = |a| = a$
- $D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$
- $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$
- $D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ where C_{ij} is the cofactor of a_{ij} .
- If we interchange two rows or two columns of a determinant then value of the determinant remains same only sign changes.
- If we add a constant multiple of a row or column to any row or column of a determinant then the value of determinant remains same.
- If we multiply a constant k to all the elements of a row or column of a determinant then the value of the determinant changes by k times.
- The determinant of a matrix and its transpose are same.
- If all the elements of a row or column are zeroes then the value of determinant is zero.
- If any two rows or two columns are identical or proportional then the value of the determinant is zero.

• **Product of Determinants:**

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} d_1 & e_1 & f_1 \\ d_2 & e_2 & f_2 \\ d_3 & e_3 & f_3 \end{vmatrix} = \begin{vmatrix} a_1d_1 + a_1e_1 + a_1f_1 & a_1d_2 + b_1e_2 + c_1f_2 & a_1d_3 + b_1e_3 + c_1f_3 \\ a_2d_1 + b_2e_1 + c_2f_1 & a_2d_2 + b_2e_2 + c_2f_2 & a_2d_3 + b_2e_3 + c_2f_3 \\ a_3d_1 + b_3e_1 + c_3f_1 & a_3d_2 + b_3e_2 + c_3f_2 & a_3d_3 + b_3e_3 + c_3f_3 \end{vmatrix}.$$

- **Conjugate of a determinant:** If $a_i, b_i, c_i \in C(i=1,2,3)$,

$$\text{If } Z = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ then, } \bar{Z} = \begin{vmatrix} \bar{a}_1 & \bar{b}_1 & \bar{c}_1 \\ \bar{a}_2 & \bar{b}_2 & \bar{c}_2 \\ \bar{a}_3 & \bar{b}_3 & \bar{c}_3 \end{vmatrix}.$$

- **Determinant of Triangular Matrix:** - The determinant of a triangular matrix is same as their product of the diagonal elements.

- **Summation of determinants:** Let $A_r = \begin{vmatrix} f(r) & a & d \\ g(r) & b & e \\ h(r) & c & f \end{vmatrix}$, then $\sum_{r=1}^n A_r = \begin{vmatrix} \sum_{r=1}^n f(r) & a & d \\ \sum_{r=1}^n g(r) & b & e \\ \sum_{r=1}^n h(r) & c & f \end{vmatrix}$, here

function of r can be elements of only one row or one column.

- **Differentiation of a Determinants:**

- (1) Let $A(x)$ be the determinant of order two. If $A(x) = \begin{vmatrix} C_1 & C_2 \end{vmatrix}$, where C_1, C_2 denotes the first column and second column, then :

$$A'(x) = \begin{vmatrix} C_1' & C_2 \end{vmatrix} + \begin{vmatrix} C_1 & C_2' \end{vmatrix}.$$

- (2) Let $A(x)$ be the determinant of order two. If $A(x) = \begin{vmatrix} R_1 \\ R_2 \end{vmatrix}$, where R_1, R_2 denotes the first row and second row, then :

$$A'(x) = \begin{vmatrix} R_1' \\ R_2 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2' \end{vmatrix}$$

- (3) Let $A(x)$ be the determinant of order two. If $A(x) = \begin{vmatrix} C_1 & C_2 & C_3 \end{vmatrix}$, where C_1, C_2, C_3 denotes the first second and third column respectively, then :

$$A'(x) = \begin{vmatrix} C_1' & C_2 & C_3 \end{vmatrix} + \begin{vmatrix} C_1 & C_2' & C_3 \end{vmatrix} + \begin{vmatrix} C_1 & C_2 & C_3' \end{vmatrix}.$$

- (4) Let $A(x)$ be the determinant of order two. If $A(x) = \begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix}$, where R_1, R_2, R_3 denotes

the first, second and third row respectively, then :

$$A'(x) = \begin{vmatrix} R_1' \\ R_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2' \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2 \\ R_3' \end{vmatrix}.$$

- (5) If only one row or one column consists of x and other rows or columns are constants,

$$\text{i.e. } A(x) = \begin{vmatrix} f(x) & a & d \\ g(x) & b & e \\ h(x) & c & f \end{vmatrix}, \text{ then}$$

$$\Rightarrow A'(x) = \begin{vmatrix} f'(x) & a & d \\ g'(x) & b & e \\ h'(x) & c & f \end{vmatrix} \quad \text{and} \quad \Rightarrow A''(x) = \begin{vmatrix} f''(x) & a & d \\ g''(x) & b & e \\ h''(x) & c & f \end{vmatrix}.$$

- **Integration of Determinants:** Let $A(x) = \begin{vmatrix} f(x) & a & d \\ g(x) & b & e \\ h(x) & c & f \end{vmatrix}$, then

$$\Rightarrow \int_r^s A(x)dx = \begin{vmatrix} \int_r^s f(x)dx & a & d \\ \int_r^s g(x)dx & b & e \\ \int_r^s h(x)dx & c & f \end{vmatrix}$$

Note: If more than one row or column of determinant contains functions of x then integration can be evaluated after evaluating the determinants.

- **Symmetric Determinants:** A determinant is called as symmetric if $a_{ij} = a_{ji}$, $\forall i, j$

Example: $\begin{vmatrix} a & b & c \\ b & d & e \\ c & e & f \end{vmatrix}$.

- **Skew Symmetric Determinants:** A determinant is called as skew symmetric if $a_{ij} = -a_{ji}$, $\forall i, j$.

Example: $\begin{vmatrix} 0 & b & c \\ -b & 0 & e \\ -c & -e & 0 \end{vmatrix}$

- (1) Every diagonal elements of skew symmetric determinants is always zero.
- (2) The value of skew symmetric determinants of even order is always a perfect square and that of odd order is always zero.

- $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ its **minor** are $M_{11} = a_{22}, M_{12} = a_{21}, M_{21} = a_{12}, M_{22} = a_{11}$

- $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ Its minors are $\rightarrow$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}, M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$M_{31} = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}, M_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}, M_{33} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

- **Cofactors of Elements of a Determinant:** - Cofactors $C_{ij} = (-1)^{i+j} M_{ij}$.
- **Solution of system of non-homogeneous linear equation by Cramer's Rule:** -
For Two Variable: -

Solve $a_{11}x + a_{12}y = b_1$ and $a_{21}x + a_{22}y = b_2$

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad D_x = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad D_y = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$$

Conditions: -

1. If $D \neq 0$, then it has unique solution $x = \frac{D_x}{D}$ and $y = \frac{D_y}{D}$.
2. If $D = D_x = D_y = 0$, then it has infinite many solution.
3. If $D = 0$, but one of the D_x, D_y is non zero then it has no solution.

For Three Variable: -

Solve $a_{11}x + a_{12}y + a_{13}z = b_1$, $a_{21}x + a_{22}y + a_{23}z = b_2$ and $a_{31}x + a_{32}y + a_{33}z = b_3$

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \quad D_x = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix},$$

$$D_y = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad D_z = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

Conditions: -

1. If $D \neq 0$, then it has unique solution $x = \frac{D_x}{D}$, $y = \frac{D_y}{D}$ and $z = \frac{D_z}{D}$.
 2. If $D = D_x = D_y = D_z = 0$, then it has infinite many solution.
 3. If $D = 0$, but one of the D_x, D_y, D_z is non zero then it has no solution.
- **Solution of system of non-homogeneous linear equation by Matrix Method:** -

For Two Variable: -

Solve $a_{11}x + a_{12}y = b_1$ and $a_{21}x + a_{22}y = b_2$

Its matrix form is $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \Rightarrow AX = B,$

Where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$.

Here $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is called as Coefficient matrix

Conditions: -

1. If $|A| \neq 0$, then it has unique solution $X = A^{-1}B$.
2. If $|A| = 0$, $Adj(A) = 0$ then it has infinite many solution.
3. If $|A| = 0$, $Adj(A) \neq 0$ then it has no solution.

For Three Variable: -

Solve $a_{11}x + a_{12}y + a_{13}z = b_1$, $a_{21}x + a_{22}y + a_{23}z = b_2$ and $a_{31}x + a_{32}y + a_{33}z = b_3$

Its matrix form is $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow AX = B$

Where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Here $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is called as coefficient matrix

Conditions: -

1. If $|A| \neq 0$, then it has unique solution $X = A^{-1}B$.
2. If $|A| = 0$, $Adj(A) = 0$ then it has infinite many solution.
3. If $|A| = 0$, $Adj(A) \neq 0$ then it has no solution.

- **Solution of system of homogeneous linear equation by Matrix Method: -**

For Two Variable: - (For Homogeneous equation solution must exist).

Solve $a_{11}x + a_{12}y = 0$ and $a_{21}x + a_{22}y = 0$

Its matrix form is $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow AX = B$,

Where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Here $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is called as Coefficient matrix

Conditions: -

1. If $|A| \neq 0$, then it has trivial solution $x = 0, y = 0$.
2. If $|A| = 0$, then it has non-trivial solution or infinite many solution.

For Three Variable: -

Solve $a_{11}x + a_{12}y + a_{13}z = 0$, $a_{21}x + a_{22}y + a_{23}z = 0$ and $a_{31}x + a_{32}y + a_{33}z = 0$

Its matrix form is $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow AX = B$

Where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Here $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is called as coefficient matrix

Conditions: - (For Homogeneous equation solution must exist).

1. If $|A| \neq 0$, then it has unique solution $x = 0, y = 0, z = 0$.
2. If $|A| = 0$, then it has non trivial solution or infinite many solution.

- **Rank of a matrix:** The rank of a matrix is the order of the highest order non singular square matrix.
- Rank of a matrix can be found by reducing the system of equation into Echelon form.
 1. Rank of a matrix in Echelon form is equal to the number of non zero rows of the matrix.
 2. The rank of a unit matrix of order $n \times n$ is n .
 3. The rank of non singular matrix of order $n \times n$ is n .
- Let $AX = B$ be a system of equations in n unknowns, such that the rank of the co-efficient matrix A is r_1 and the rank of the augmented matrix K is r_2 .
 1. If $r_1 \neq r_2$, then the system $AX = B$ is inconsistent. i.e. it has no solution.
 2. If $r_1 = r_2 = n$, then the system $AX = B$ is consistent and has unique solution.
 3. If $r_1 = r_2 < n$, then the system $AX = B$ is consistent and has many solution.
- Let system of equations are :

$$a_{11}x + a_{12}y + a_{13}z = b_1, a_{21}x + a_{22}y + a_{23}z = b_2 \text{ and } a_{31}x + a_{32}y + a_{33}z = b_3$$

Its matrix form is
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow AX = B$$

Its coefficient matrix is
$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Its Augmented matrix is
$$A = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

Its echelon form can be obtained by applying elementary row transformation to its augmented matrix to get upper triangular matrix form.

- **Rank of the Matrix by Determinant Method:** -A $m \times n$ matrix A has rank $r \geq 1$ if and only if A has $r \times r$ sub-matrix with non zero determinant, whereas the determinant of every square matrix of order $r + 1$ or more is zero.
- A square matrix $n \times n$ has rank n if $\det(A)$ is non zero.
- If $\text{rank}(A) = \text{rank}[A|B] < \text{No. of Variables}$. Then system of equation $Ax = B$ has infinite many solution.
- The sum of the products of the element of any row (column) of a square matrix with the cofactors of the corresponding elements is zero.
- If the element of a square matrix are polynomial in x and two rows (columns) become identical when $x = a$ then $x - a$ is a factor of the determinant. If three rows (columns) are identical then $(x - a)^2$ is a factor.
- $|AB| = |A| \times |B| = |BA|$
- If $|AB| = 0$ then either $|A| = 0$ or $|B| = 0$.
- The determinant of a skew symmetric matrix of order 3 is 0.
- The determinant of a unit matrix is 1.
- A square matrix is said to be **singular** if $|A| = 0$ and **non-singular** if $|A| \neq 0$.
- If A is a square matrix of order $n \times n$, then $|kA| = k^n |A|$.
- If A is a scalar matrix of order $n \times n$ with all the diagonal elements are k, then $|A| = k^n$.
- $|A| = |A^T|$
- $$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc).$$
- $$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2.$$

$$\bullet \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a).$$

$$\bullet \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c).$$

$$\bullet \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\bullet \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a).$$

$$\bullet \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca).$$

$$\bullet \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & b^3 \end{vmatrix} = abc(a-b)(b-c)(c-a).$$

$$\bullet \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c.$$

$$\bullet \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right).$$

$$\bullet \begin{vmatrix} 1+a^2 & ab & ac \\ ab & 1+b^2 & bc \\ ac & bc & 1+c^2 \end{vmatrix} = 1+a^2+b^2+c^2.$$
